

# Induced Bipartite Subgraphs, Average Eccentricity, and Hamiltonian Paths

A structural proof of WOWII Conjecture 198a

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## Abstract

Let  $b(G)$  denote the maximum order of an induced bipartite subgraph of a finite connected simple graph  $G$ , and let  $\overline{\text{ecc}}(G)$  denote its average eccentricity. We give a self-contained proof that  $b(G) \leq 2 + \overline{\text{ecc}}(G)$  implies the existence of a Hamiltonian path, the assertion of Graffiti.pc Conjecture 198a in *Written on the Wall II*. The proof reduces the problem to two extremal possibilities relative to the diameter  $D$ . When  $b(G) = D + 1$ , the vertices outside a diametral geodesic admit a partition into cliques that can be inserted along that geodesic. In the remaining case the graph is self-centred; distance-layer constructions exclude  $D \geq 3$ , and a longest-path argument settles  $D = 2$ . The argument also yields structural statements independent of the average-eccentricity hypothesis. We show that the additive constant 2 is optimal, and describe the accompanying Lean development together with the precise status of its reported verification.

**Keywords.** Hamiltonian path; induced bipartite subgraph; average eccentricity; diameter; self-centred graph; formal verification.

## 1 Introduction

A Hamiltonian path in a finite graph visits every vertex exactly once. A graph admitting such a path is called *traceable*. We study a sufficient condition for traceability that compares an induced-subgraph invariant with an average metric invariant. For a graph  $G = (V, E)$ , define

$$b(G) = \max\{|S| : S \subseteq V \text{ and } G[S] \text{ is bipartite}\}.$$

Here the bipartite subgraph is required to be *induced*; all edges of  $G$  between vertices of  $S$  are retained. If  $G$  is connected and  $n = |V|$ , set

$$\text{ecc}_G(v) = \max_{u \in V} d_G(u, v), \quad \overline{\text{ecc}}(G) = \frac{1}{n} \sum_{v \in V} \text{ecc}_G(v).$$

The following is the assertion of Conjecture 198a in DeLaVina's *Written on the Wall II* collection of Graffiti.pc conjectures [1, 2].

**Theorem 1.1** (WOWII Conjecture 198a). *Let  $G$  be a finite connected simple graph on  $n \geq 2$  vertices. If*

$$b(G) \leq 2 + \overline{\text{ecc}}(G), \tag{1}$$

*then  $G$  has a Hamiltonian path.*

The proof exploits the narrow interval available for  $b(G)$  under (1). A diametral geodesic gives  $b(G) \geq D+1$ , whereas  $\overline{\text{ecc}}(G) \leq D$  gives  $b(G) \leq D+2$ . Equality at the lower endpoint forces a clique structure around the geodesic. Equality at the upper endpoint forces every vertex to have eccentricity  $D$ . The latter condition is incompatible with  $b(G) = D+2$  once  $D \geq 3$ . At diameter two, the obstruction to extending a longest path produces an induced bipartite subgraph on five vertices.

**Related work and scope.** A public proof announcement for the same conjecture was posted by the GitHub user `lukekabbash` on July 23, 2026, with links to a manuscript and a Lean artifact [3, 4]. The present article gives an exposition of the supplied geodesic-and-longest-path argument and makes no claim of priority. In particular, an “open” label in a conjecture repository is not by itself evidence that no proof has been announced. The mathematical proof below is complete without recourse to a proof assistant. The reported Lean compilation and the separate, still-pending comparison with the repository theorem are distinguished in section 7.

## 2 Geodesics and the numerical reduction

All graphs in the sequel are finite and simple. Unless otherwise stated,  $G$  is connected,  $n \geq 2$ , and  $D = \text{diam}(G)$ . We write  $x \sim y$  for adjacency and suppress the subscript  $G$  on distances and eccentricities. A vertex is *dominating* if it is adjacent to every other vertex. A graph is *self-centred* if every vertex has eccentricity equal to its diameter. Paths are written as sequences of distinct vertices; their lengths count edges.

Fix a diametral geodesic

$$P = v_0 v_1 \cdots v_D, \quad d(v_0, v_D) = D.$$

Every subpath of  $P$  is a shortest path, so

$$d(v_i, v_j) = |i - j| \quad (0 \leq i, j \leq D). \quad (2)$$

Consequently  $G[V(P)]$  is an induced path. If a vertex  $x$  is adjacent to both  $v_i$  and  $v_j$ , then

$$|i - j| = d(v_i, v_j) \leq 2. \quad (3)$$

Thus the neighbours of any vertex on  $P$  lie within a window of at most three consecutive indices.

We repeatedly certify bipartiteness by assigning integer labels  $f$  to a vertex set  $S$  such that every edge of  $G[S]$  joins vertices with labels of opposite parity. The two parity classes then form a bipartition, and  $|S| \leq b(G)$ .

**Lemma 2.1** (Extremal reduction). *If  $G$  satisfies (1), then*

$$b(G) \in \{D+1, D+2\}.$$

*If  $b(G) = D+2$ , then  $G$  is self-centred.*

*Proof.* The induced path  $G[V(P)]$  is bipartite and has  $D+1$  vertices. Since every eccentricity is at most  $D$ , we have

$$D+1 \leq b(G) \leq 2 + \overline{\text{ecc}}(G) \leq D+2. \quad (4)$$

The first assertion follows because  $b(G)$  is an integer. If  $b(G) = D + 2$ , all inequalities on the right of (4) are equalities, so

$$\sum_{v \in V} (D - \text{ecc}(v)) = 0.$$

Every summand is nonnegative; hence  $\text{ecc}(v) = D$  for every  $v \in V$ .  $\square$

### 3 The extremal case $b(G) = D + 1$

The lower endpoint of (4) already forces traceability, without any assumption on average eccentricity.

**Proposition 3.1** (Clique insertion along a geodesic). *Let  $G$  be a finite connected simple graph of diameter  $D \geq 1$ . If  $b(G) = D + 1$ , then  $G$  has a Hamiltonian path.*

*Proof.* Fix the diametral geodesic  $P = v_0 \cdots v_D$ . We first prove that each vertex  $x \notin V(P)$  is adjacent to two consecutive vertices of  $P$ . Otherwise, by (3), all indices of neighbours of  $x$  on  $P$  have the same parity: two indices of different parity would differ by one. Colour  $v_i$  by  $i \bmod 2$  and give  $x$  the opposite colour to its neighbours on  $P$ . If it has no such neighbour, either colour works. This is a bipartition of  $G[V(P) \cup \{x\}]$ , contradicting  $b(G) = D + 1$ .

For  $x \notin V(P)$ , define its *slot* by

$$s(x) = \min\{k : 0 \leq k < D, x \sim v_k, x \sim v_{k+1}\}.$$

If  $s(x) = k$ , then

$$N_G(x) \cap V(P) \subseteq \{v_k, v_{k+1}, v_{k+2}\} \cap V(P), \quad (5)$$

where indices outside  $\{0, \dots, D\}$  are omitted. Indeed, applying (3) to the neighbours  $v_k, v_{k+1}$  shows that any further neighbour has index between  $k - 1$  and  $k + 2$ . If  $k \geq 1$ , adjacency to  $v_{k-1}$  would contradict the minimality of  $k$ .

For  $0 \leq k < D$ , let

$$C_k = \{x \in V \setminus V(P) : s(x) = k\}.$$

We claim that  $C_k$  is a clique. If distinct  $x, y \in C_k$  were nonadjacent, consider

$$S = (V(P) \setminus \{v_{k+1}\}) \cup \{x, y\}.$$

Colour each remaining  $v_i$  by  $i \bmod 2$ , and colour both  $x$  and  $y$  by  $(k + 1) \bmod 2$ . By (5), their remaining neighbours on  $P$  can only be  $v_k$  and  $v_{k+2}$ , both of the opposite colour. There is no edge  $xy$ , and all edges between remaining vertices of  $P$  are properly coloured. Thus  $G[S]$  is bipartite, but

$$|S| = (D + 1) - 1 + 2 = D + 2,$$

again contradicting the hypothesis.

The sets  $C_0, \dots, C_{D-1}$  partition  $V \setminus V(P)$ . List each  $C_k$  in any order, omitting empty sets. The sequence

$$v_0, C_0, v_1, C_1, \dots, v_{D-1}, C_{D-1}, v_D \quad (6)$$

contains every vertex exactly once. Consecutive vertices within each block are adjacent because the block is a clique, and every vertex of  $C_k$  is adjacent to both  $v_k$  and  $v_{k+1}$ . An empty block leaves the edge  $v_k v_{k+1}$ . Hence (6) is a Hamiltonian path.  $\square$

## 4 Induced bipartite sets in self-centred graphs

The upper endpoint of (4) leads to a different phenomenon: eccentric vertices allow a diametral geodesic to be enlarged while preserving an explicit bipartition.

**Proposition 4.1.** *Let  $G$  be a finite connected self-centred graph of diameter  $D \geq 3$ . Then  $b(G) \geq D + 3$ . More precisely, if  $D \geq 4$ , then*

$$b(G) \geq D + 1 + \left\lfloor \frac{D}{2} \right\rfloor. \quad (7)$$

*Proof.* Fix a diametral geodesic  $P = v_0 \cdots v_D$ .

*Case 1:*  $D \geq 4$ . Put  $m = \lfloor D/2 \rfloor$ ,  $r = \lceil D/2 \rceil$ , and  $c = v_m$ . Self-centredness gives a vertex  $w$  with  $d(c, w) = D$ . Choose a geodesic

$$Q = q_0 q_1 \cdots q_D, \quad q_0 = c, \quad q_D = w.$$

Since  $d(c, q_j) = j$  and  $d(c, v_i) = |i - m| \leq r$ , the sets  $V(P)$  and  $\{q_{r+1}, \dots, q_D\}$  are disjoint. Set

$$S = V(P) \cup \{q_{r+1}, \dots, q_D\}, \quad f(x) = d(c, x).$$

For every edge  $xy$ , the triangle inequality gives  $|f(x) - f(y)| \leq 1$ . We show that equality of the two labels is impossible for an edge of  $G[S]$ . If both endpoints lie on  $P$ , they have consecutive indices, and the integers  $|i - m|$  and  $|i + 1 - m|$  differ by one. If both lie in the selected tail of  $Q$ , distinct vertices have distinct distances from  $c$ . Finally, a vertex of  $P$  has label at most  $r$ , whereas a selected vertex of  $Q$  has label greater than  $r$ . Thus every edge of  $G[S]$  joins opposite parities, proving bipartiteness. Counting vertices gives

$$|S| = D + 1 + (D - r) = D + 1 + \left\lfloor \frac{D}{2} \right\rfloor \geq D + 3.$$

*Case 2:*  $D = 3$ . Choose vertices  $w_1, w_2$  with

$$d(v_1, w_1) = 3, \quad d(v_2, w_2) = 3.$$

Both lie outside  $V(P)$ , since every vertex of  $P$  is within distance two of each of  $v_1, v_2$ . The distance conditions imply

$$N(w_1) \cap V(P) \subseteq \{v_3\}, \quad N(w_2) \cap V(P) \subseteq \{v_0\}. \quad (8)$$

For example, adjacency of  $w_1$  to  $v_0, v_1$ , or  $v_2$  would give a  $v_1$ – $w_1$  path of length at most two.

If  $w_1 \neq w_2$ , the graph induced by  $V(P) \cup \{w_1, w_2\}$  is a subgraph of the cycle

$$v_0 v_1 v_2 v_3 w_1 w_2 v_0.$$

Some displayed edges may be absent, but (8) and the inducedness of  $P$  exclude every other edge. It is therefore bipartite and has six vertices.

Suppose now that  $w_1 = w_2 = w$ . By (8),  $w$  has no neighbour on  $P$ . Choose a geodesic  $v_1, a, \beta, w$ . Then  $d(v_1, \beta) = 2$  and  $\beta \sim w$ . The latter adjacency shows that  $\beta \notin V(P)$ . Moreover,  $\beta \not\sim v_1$ , and  $\beta \not\sim v_2$ , since otherwise  $d(v_2, w) \leq 2$ . It cannot be adjacent to both  $v_0$  and  $v_3$ , because that would imply  $d(v_0, v_3) \leq 2$ . Consequently  $G[V(P) \cup \{\beta, w\}]$  consists of the induced path  $P$ , the edge  $\beta w$ , and at most one edge joining  $\beta$  to an endpoint of  $P$ . It is a forest, possibly disconnected, on six distinct vertices. Thus  $b(G) \geq 6 = D + 3$  in this case as well.  $\square$

*Remark 4.2.* The distance-layer construction uses only the existence of an eccentric vertex at distance  $D$  from the chosen midpoint  $v_{\lfloor D/2 \rfloor}$ . Full self-centredness is a convenient sufficient hypothesis. At  $D = 3$ , the bound  $b(G) \geq D + 3$  is attained by the cycle  $C_6$ .

## 5 The diameter-two case

It remains to handle a self-centred graph of diameter two with  $b(G) = 4$ . The following statement isolates exactly what is needed.

**Proposition 5.1.** *Let  $G$  be a finite connected simple graph of diameter two. If  $G$  has no dominating vertex and  $b(G) \leq 4$ , then  $G$  has a Hamiltonian path.*

*Proof.* Let  $L = p_0 p_1 \cdots p_{k-1}$  be a longest path. Suppose, towards a contradiction, that  $h \notin V(L)$ . Let  $H$  be the vertex set of the component of  $G - V(L)$  containing  $h$ , and let

$$A = \{p_i \in V(L) : N(p_i) \cap H \neq \emptyset\}$$

be the set of attachment vertices. Whenever  $p_i, p_j \in A$ , there exists a  $p_i$ - $p_j$  path whose internal vertices all lie in  $H$ : choose one neighbour in  $H$  of each endpoint and join them by a path in the connected graph  $G[H]$ . If the neighbours coincide, the path has just that one internal vertex.

Maximality of the order of  $L$  gives the following four restrictions.

- (i) Neither  $p_0$  nor  $p_{k-1}$  belongs to  $A$ . Otherwise a neighbour in  $H$  extends the corresponding end of  $L$ .
- (ii) If  $p_i \in A$ , then  $p_{i+1}$  has no neighbour in  $H$ . The successor exists by (i). If it also had a neighbour in  $H$ , replacing the edge  $p_i p_{i+1}$  by a path through  $H$  would give a longer path containing all vertices of  $L$ .
- (iii) If  $p_i \in A$ , then  $p_0 \not\sim p_{i+1}$ . Indeed, if  $z \in H$  is adjacent to  $p_i$  and  $p_0 \sim p_{i+1}$ , then

$$z, p_i, p_{i-1}, \dots, p_0, p_{i+1}, p_{i+2}, \dots, p_{k-1}$$

is a longer path. Here  $i \geq 1$  by (i).

- (iv) If  $p_i, p_j \in A$  and  $i < j$ , then  $p_{i+1} \not\sim p_{j+1}$ . To see this, let  $p_i, z_1, \dots, z_t, p_j$  be a path with  $z_1, \dots, z_t \in H$  and  $t \geq 1$ . An edge  $p_{i+1} p_{j+1}$  would give the path

$$p_0, \dots, p_i, z_1, \dots, z_t, p_j, p_{j-1}, \dots, p_{i+1}, p_{j+1}, \dots, p_{k-1}.$$

The listed segments of  $L$  are disjoint and together contain all of  $V(L)$ , so this again contradicts maximality. Consecutive attachment vertices are already excluded by (ii).

We next show that  $|A| \geq 2$ . For each  $z \in H$ , restriction (i) and the diameter-two hypothesis give  $d(z, p_0) = 2$ . Let  $y$  be a common neighbour of  $z$  and  $p_0$ . If  $y \notin V(L)$ , the edge  $zy$  places  $y$  in the same component  $H$ , contradicting (i). Therefore  $y \in A$ , and in particular  $A$  is nonempty.

Suppose  $A = \{s\}$ . The same argument, applied to every  $z \in H$ , shows that every vertex of  $H$  is adjacent to  $s$ . There are no edges from  $H$  to  $V \setminus (H \cup \{s\})$ : an endpoint on  $L$  would be another attachment vertex, and an endpoint outside  $L$  would belong to the component  $H$ . Now fix  $w \in V \setminus (H \cup \{s\})$ . Because  $h \not\sim w$  and  $D = 2$ , there is a common neighbour  $y$  of  $h$  and  $w$ . Such a  $y$  cannot lie in  $H$ , by the absence of edges just established. Since every neighbour of  $h$  outside  $H$  must be  $s$ , it follows that  $y = s$  and hence  $s \sim w$ . Thus  $s$  is adjacent to all other vertices, contrary to the absence of a dominating vertex. This proves  $|A| \geq 2$ .

Choose  $p_i, p_j \in A$  with  $i < j$ . By (i),  $1 \leq i$  and  $j + 1 < k$ . The set

$$I = \{h, p_0, p_{i+1}, p_{j+1}\}$$

has four distinct vertices and is independent. Indeed,  $h$  has no edge to  $p_0$  by (i) and no edge to the two successors by (ii); the remaining three possible edges are ruled out by (iii) and (iv). Also  $p_i \notin I$ . Therefore  $G[I \cup \{p_i\}]$  is bipartite, with bipartition  $(I, \{p_i\})$ , and has five vertices. This contradicts  $b(G) \leq 4$ . Hence  $L$  is Hamiltonian.  $\square$

*Remark 5.2.* The assumption excluding a dominating vertex cannot simply be dropped: the star  $K_{1,3}$  has diameter two and  $b(K_{1,3}) = 4$ , but has no Hamiltonian path.

## 6 Proof of the main theorem and optimality

*Proof of theorem 1.1.* By lemma 2.1, either  $b(G) = D + 1$  or  $b(G) = D + 2$ . In the first case, proposition 3.1 gives a Hamiltonian path.

In the second case, lemma 2.1 shows that  $G$  is self-centred. Since  $G$  is connected and has at least two vertices,  $D \geq 1$ . If  $D = 1$ , the graph is complete and any ordering of its vertices is a Hamiltonian path. If  $D = 2$ , every vertex has eccentricity two, so no vertex is dominating; since  $b(G) = 4$ , proposition 5.1 applies. Finally, if  $D \geq 3$ , proposition 4.1 gives  $b(G) \geq D + 3$ , contradicting  $b(G) = D + 2$ . These cases exhaust the possibilities.  $\square$

The diameter-one subcase of the upper-endpoint alternative is in fact empty: a complete graph of order at least two has  $b(G) = 2 = D + 1$ . It was included above to keep the diameter split explicit.

The coefficient-free additive threshold in theorem 1.1 is best possible, as the following family shows.

**Proposition 6.1** (Optimality of the additive constant). *For every real  $c > 2$ , there exists a finite connected nontraceable simple graph  $G$  such that  $b(G) \leq c + \overline{\text{ecc}}(G)$ .*

*Proof.* For an integer  $t \geq 1$ , let  $G_t$  consist of a clique  $X$  of order  $t$ , two further vertices  $a, b$ , and a vertex  $u$  adjacent to every vertex of  $X \cup \{a, b\}$ . Apart from the clique edges and the edges incident with  $u$ , there are no edges. Thus  $a$  and  $b$  are leaves and  $|V(G_t)| = t + 3$ .

Deleting  $u$  leaves three nonempty components, namely  $X$ ,  $\{a\}$ , and  $\{b\}$ . Deleting a vertex from a Hamiltonian path leaves at most two path components, whose edges must lie in  $G_t - u$ . Those two paths cannot cover three components of  $G_t - u$ . Consequently  $G_t$  is not traceable.

The vertices  $u, a, b$  and any  $x \in X$  induce a copy of  $K_{1,3}$ , so  $b(G_t) \geq 4$ . Conversely, a bipartite induced subgraph containing  $u$  contains at most one vertex of  $X$ , since two would form a triangle with  $u$ . Such a subgraph has order at most four. One omitting  $u$  contains at most two vertices of  $X$ , and again has order at most four. Hence  $b(G_t) = 4$ .

The vertex  $u$  has eccentricity one, and every other vertex has eccentricity two. It follows that

$$\overline{\text{ecc}}(G_t) = \frac{1 + 2(t + 2)}{t + 3} = 2 - \frac{1}{t + 3}, \quad b(G_t) - \overline{\text{ecc}}(G_t) = 2 + \frac{1}{t + 3}.$$

Choose  $t$  large enough that  $1/(t + 3) \leq c - 2$ . Then  $b(G_t) \leq c + \overline{\text{ecc}}(G_t)$ , as required.  $\square$

## 7 Lean formalization and verification status

The accompanying source development is organised into five Lean files under the namespace `WOW198a`. Its mathematical decomposition matches the proof above.

| Module                  | Mathematical content                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>Basic.lean</code> | Parity certificates for induced bipartite sets; geodesic distance identities; Hamiltonian walks from vertex lists. |
| <code>L1.lean</code>    | Slot construction, clique blocks, and the Hamiltonian path in <a href="#">proposition 3.1</a> .                    |
| <code>L2.lean</code>    | The distance-layer construction and the separate diameter-three argument of <a href="#">proposition 4.1</a> .      |
| <code>L3.lean</code>    | Longest-path modifications and the diameter-two criterion of <a href="#">proposition 5.1</a> .                     |
| <code>Main.lean</code>  | Numerical reduction and assembly of the main theorem.                                                              |

The graph invariants are imported from `FormalConjecturesForMathlib`. Hamiltonian walks use Mathlib’s corresponding API. The endpoint is `WOW198a.conjecture198a`. It assumes a finite nontrivial vertex type, a connected simple graph, and the inequality between `b G` and `2 + averageEccentricity G`. Its conclusion is the existence of vertices  $a, b$  and a walk from  $a$  to  $b$  satisfying `p.IsHamiltonian`. The `Nontrivial` assumption expresses  $n \geq 2$ . For a finite connected graph, the extended-natural eccentricities used in the library are finite and agree with the ordinary integer eccentricities in this article.

The supplied development reports successful compilation with Lean 4.33.1 and Mathlib v4.33.1, without `sorry`, and reports that

```
#print axioms WOW198a.conjecture198a
```

returns exactly

```
[propext, Classical.choice, Quot.sound]
```

These are the standard logical axioms reported for this proof; in particular, the reported list contains no `sorryAx`. The five supplied source files contain no proof-placeholder `sorry` declarations.

**Scope of the verification claim.** The compilation and axiom output above are the reported results accompanying the supplied proof, rather than a fresh build performed for this manuscript. According to the accompanying progress report, the final kernel-level type comparison with

```
WrittenOnTheWallIII.GraphConjecture198a.conjecture198a
```

was still awaiting completion of the full Mathlib build. This article therefore does not assert that the separate comparison has already passed. The theorem signatures agree on inspection, but that inspection does not replace the pending check in the intended repository environment. No pinned checkout or completed comparison log was included with the supplied source package.

For reproducibility, the package’s README instructs the reader to copy `WOW198a/` into a compatible `formal-conjectures` checkout, register the `WOW198a` library in `lakefile.toml`, and run

```
lake build WOW198a.Main
```

The optimality statement in [proposition 6.1](#) is proved here and is not part of the supplied Lean development. Equation (7) records the full size of the constructed set; the formal lemma exports the weaker bound  $D + 3$  used in the main theorem.

## References

- [1] E. DeLaVina, *Written on the Wall II: Conjectures of Graffiti.pc*, Conjecture 198a (2006). <http://cms.uhd.edu/faculty/delavinae/research/wowII/>. Original collection; statement also reproduced in [2].
- [2] The Formal Conjectures Authors, *Written on the Wall II—Conjecture 198a*, `google-deepmind/formal-conjectures`, Lean source file `GraphConjecture198a.lean`. GitHub source file. Consulted September 23, 2026.
- [3] `lukekabbash`, *A Lean 4 proof of WOWII Conjecture 198a*, public proof and verification repository, July 2026. <https://github.com/lukekabbash/wowii-198a-lean>. Consulted September 23, 2026.
- [4] `lukekabbash`, *Written on the Wall II, Conjecture 198a*, issue #4596 in `google-deepmind/formal-conjectures`, opened July 23, 2026. <https://github.com/google-deepmind/formal-conjectures/issues/4596>. Consulted September 23, 2026.